ENSEMBLESET: A NEW HIGH QUALITY SYNTHESISED DATASET FOR CHAMBER ENSEMBLE SEPARATION

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ABSTRACT

Music source separation research has made great advances in recent years, especially towards the problem of separating vocals, drums, and bass stems from mastered songs. The advances in this field can be directly attributed to the availability of large-scale multitrack research datasets for these mentioned stems. Tasks such as separating similar-sounding sources from an ensemble recording have seen limited research due to the lack of sizeable, bleed-free multitrack datasets. In this paper, we introduce a novel multitrack dataset called EnsembleSet generated using the Spitfire BBC Symphony Orchestra library using ensemble scores from RWC Classical Music Database and Mutopia. Our data generation method introduces automated articulation mapping for different playing styles based on the input MIDI/MusicXML data. The sample library also enables us to render the dataset with 20 different mix/microphone configurations allowing us to study various recording scenarios for each performance. The dataset presents 80 tracks (6+ hours) with a range of string, wind, and brass instruments arranged as chamber ensembles. We also present our benchmark on our synthesised dataset using a permutation-invariant time-domain separation model for chamber ensembles which produces generalisable results when tested on real recordings from existing datasets.

1. INTRODUCTION

Audio source separation aims to extract individual sound sources from a digital audio mixture. Based on the constituents of the input mixture and the target output, the problem definition can be further refined to specific audio separation tasks like speech separation, speech enhancement, and music source separation [1]. While specific subtasks in the speech-domain like speech denoising, multi-speaker separation and dereverberation have been thoroughly explored, music separation research has largely been focused on the demixing challenge [2] aided by the popular MUSDB dataset [3]. The demixing challenge is targeted at solving the problem of separation of vocals, bass and drums from mixed and mastered pop songs. This has greatly benefited the field by demonstrating that source separation is indeed possible at a commercial scale with state-of-the-art deep learning based architectures. Unfortunately, this also has resulted in the research towards this specific task to dwarf other problems that would also fall under the umbrella of music source separation, to the extent that music source separation has become synonymous with the task of separating vocal, drums and bass stems from mastered songs.

In this paper, we explore a different area in music source separation with a focus on separation of chamber ensembles, where the target sources are harmonised and have very high spectral overlap. The music demixing challenge has shown successful separation of instruments with distinct spectro-temporal cues like vocals, drums and bass. Separating monotimbral ensembles is an inherently challenging task as they combine challenging aspects of both speech and music separation. In chamber ensembles we find the sources to occupy similar frequency ranges, may have label ambiguity [4, 5] due to multiple sources belonging to the same instrument family meanwhile being temporally and harmonically correlated, due to their musical structure which further increases their spectral overlap.

To address this challenge, we present a novel chamber ensemble dataset titled EnsembleSet [6]. EnsembleSet was synthesised using a realistic sample library Spitfire BBC Symphony Orchestra (BBCSO) [7] utilising the MIDI transcriptions from the RWC Classical Music Database [8] and lilypond scores from Mutopia [9] (refer to Section 3.4 for details). In Section 4 we utilise EnsembleSet to train a source separation model based on the Dual-path Transformer architecture (DPTNet) [10, 11] for separating mixtures of chamber ensembles. We achieve very good and generalisable separation performance which we exhibit through cross-dataset performance evaluation in Section 5. Other applications of EnsembleSet may include topics such as multi-instrument transcription [12], instrument recognition [13], score-informed source separation [14], microphone translation [15], automatic mixing [16] and other tasks that may benefit from score-aligned multitrack multi-instrument data.

2. BACKGROUND

Although the term "Music Source Separation" sounds like an umbrella-term for all applications of source separation

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in music, in research the use of this term is largely used to describe the specific task of vocals/drums/bass instrument stem separation from mastered tracks [2, 17]. A music source separation task which is relatively unexplored is the challenge of separating similar sounding instruments from a mixture. This problem has two significant differences from the aforementioned task. Firstly, if the sources in the mixtures are similar sounding (e.g., mixture of a strings section), it results in high spectral energy overlap. This is further compounded by the fact that such sources often play in a very synchronised fashion while harmonising each other. Secondly, often in such mixtures we may have multiple sources of the same type present, which makes it an unsuitable problem to be solved using *class-based* separation methods [4]. We define the task of separating such mixtures with constituent sources suffering from label ambiguity and high timbral similarity as *monotimbral ensemble separation*.

2.1 Datasets

Training supervised source separation models typically requires datasets which provide the clean target sources as a reference for the deep learning models to learn from. While the majority of popular music can be recorded in separate takes for different performers, with a reference metronome or a backing track, ensembles are usually recorded together in the same take. This is due to the fact that ensemble performers rely on being able to hear each other during performance to be able to synchronise perfectly. This raises the problem that the majority of stems available from recording projects of monotimbral ensembles contain bleed [1] from non-target sources [18, 19]. This becomes problematic for training models for source separation as we do not have the clean ground truth as a target result for the model. This lack of clean and sizeable datasets for ensembles has affected the amount of research seen in this domain.

The URMP dataset [20] addresses this problem by making the performers record isolated takes and then subsequently re-align, dereverberate and downmix them to be present in a physical space. Another work [21] presented a dataset recorded by minimising the amount of bleed using soundproofing across booths while recording multiple performers in the same take. Bach10 [22] also presents multitrack recordings of chamber ensembles where each song consists of four parts (Soprano, Alto, Tenor and Bass) which were performed by violin, clarinet, saxophone and bassoon, respectively. The TRIOS dataset [23] consists of 5 bleed-free multitracks and synchronised MIDI files of 4 classical music pieces and 1 jazz piece.

2.2 Prior work

There are some tasks which fit our definition of monotimbral separation that have been explored recently. One is vocal harmony separation [11,24–26]. While the label ambiguity problem does exist for this task, some approaches have circumvented it by looking at the problem in a class-based separation fashion by categorising the constituent sources based on their registers i.e. alto, soprano, bass and tenor. One method of solving the label ambiguity problem is to tackle this problem in a score-conditioned fashion as in [25]. Another method of tackling this problem is called permutation invariant training (PIT) [27] which has been the preferred solution to tackle the label ambiguity problem for speech separation research [10, 28, 29]. PIT has been utilised for choral separation in [11]. Another approach has been to use multi-task learning by utilising score-information to simultaneously separate and transcribe mixtures of 2-source chamber ensembles which has shown some success for scenarios with small datasets [30].

3. DATASET

To overcome the challenge of bleed-free real recorded datasets for ensembles, we introduce a novel dataset “EnsembleSet", which utilises a highly realistic orchestral sample library by Spitfire Audio called "BBC Symphony Orchestra" (BBCSO) [7]. We use this sample library to render digital chamber ensemble scores from MIDI and MusicXML format to 18 unique multi-mic recordings and 2 professional mixes. For this work, we utilised the RWC Classical Music Database [8] and Mutopia [9] to source our chamber ensemble MIDI and MusicXML (converted from lilypond) scores. It must be noted that MIDI data are not ideal to capture string, wind and brass instrument scores as they do not encapsulate articulation information. On the other hand lilypond scores contain minimal dynamics (velocity) information, which is essential for realistic rendering using virtual instruments. In order to address these challenges, we utilise expression maps provided by Dorico [31], a scorewriter software which allows us to determine the articulation mode for each note in the piece.

3.1 Collecting Digital Music Scores

3.1.1 RWC Classical Music Database

The RWC Classical Music Database [8] consists of 50 public-domain classical pieces performed by musicians and then manually transcribed to MIDI with high-quality tempo and velocity mapping. Since the database only provides the final mix of these performances, its applications are limited especially in the context of source separation. We choose a subset of these pieces which contain chamber ensembles which can be rendered using our method. Our 9 selected pieces (1h 3m 34s) [2] consist of 4 string quartets, 2 clarinet quintets, 2 piano trios and 1 piano quintet. It must be noted that for the piano trios and quintet, we only render the string instrument parts. Because MIDI files lack articulation information, we modify the MIDI files using Dorico to automatically add it using keyswitches, which are then subsequently rendered as multitracks on Reaper [32].

[1] Sound picked up by a microphone from a source other than that which is intended.

[2] Rendered duration in dataset.

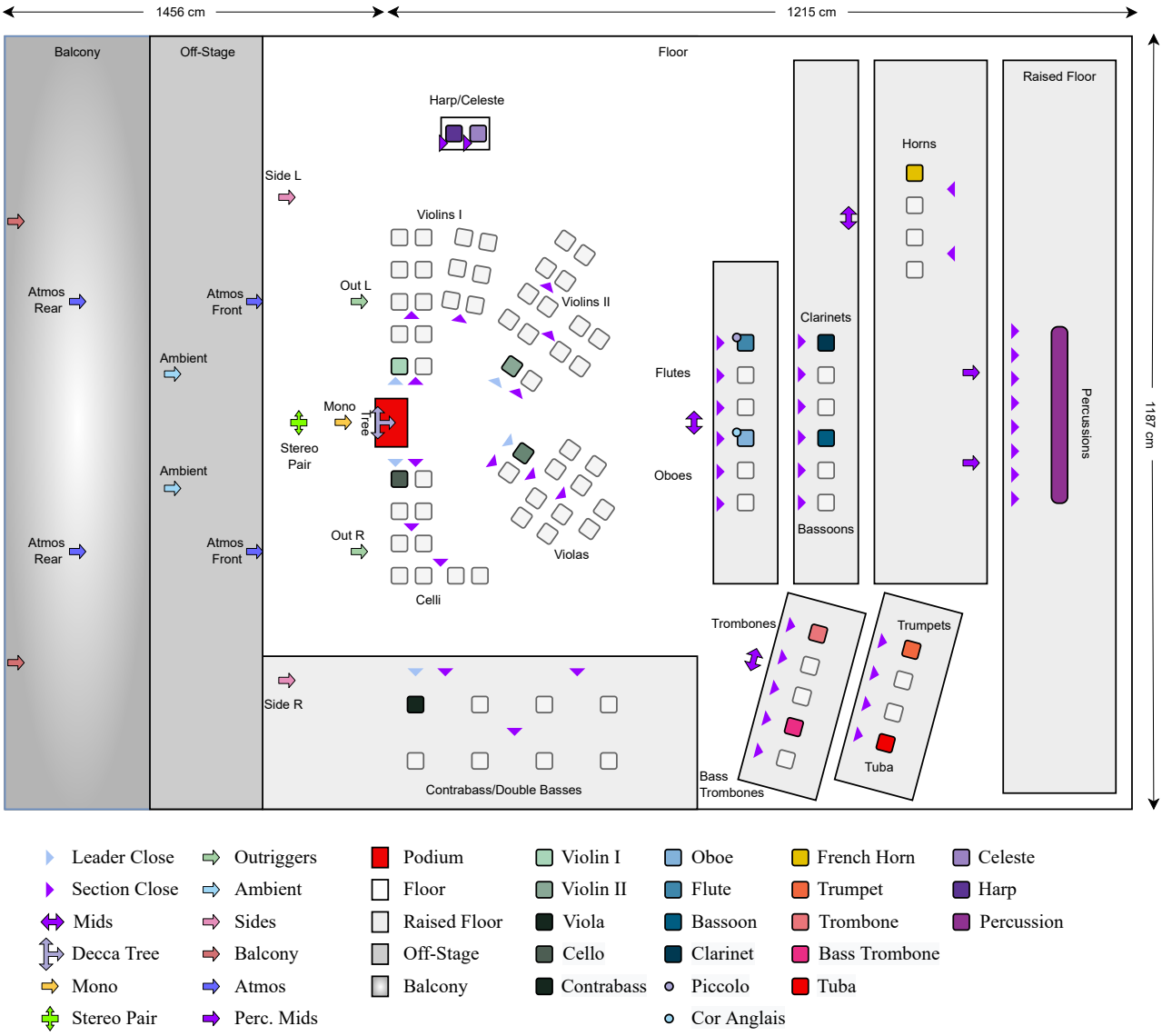


Figure 1. Recording configuration for the Spitfire Audio BBC Symphony Orchestra sample library depicting the placement of individual microphones, performers. The scale of the image is not uniform as the off-stage elements are placed further than they appear.

3.1.2 Mutopia

The Mutopia project [9] is a publicly sourced and manually verified free content sheet music library. The sheet music scores are manually annotated from old scores that are now public domain, and digitally archived using the lilypond format which can be converted to MusicXML and MIDI. This library has a large collection of string ensembles, of which we utilise 71 pieces (5h 5m 35s)[2] comprising a variety of chamber ensembles primarily composed of string quartets but also including other instruments such as Trumpet, Horn, Oboe, Clarinet, Flute and Bassoon. Although all the lilypond files come with their standard MIDI conversions, we utilise the lilypond to MusicXML conversion python library [33], to preserve the articulation information present in the lilypond files. For the files which we are able to convert to MusicXML, we import them to Dorico, where these articulations are translated to keyswitches (described in Table 2) and rendered to MIDI format which can then be utilised by the BBCSO plugin when rendered on Reaper [32].

3.1.3 Data Cleaning

Many of the scores used to render our dataset contain instruments that are absent (e.g., piano, vocals) in our sample library. Since the intent of EnsembleSet is to generate realistic renders of instruments performing and being recorded in the same physical space, we chose to remove the incompatible instruments as rendering them using other plugins will not be consistent. While converting Mutopia based files using the lilypond to MusicXML conversion tool, many files resulted in erroneous MusicXML files. For the corrupted conversions, we used the MIDI files directly from the database and were unable to preserve articulation for those pieces. For these pieces we use the same

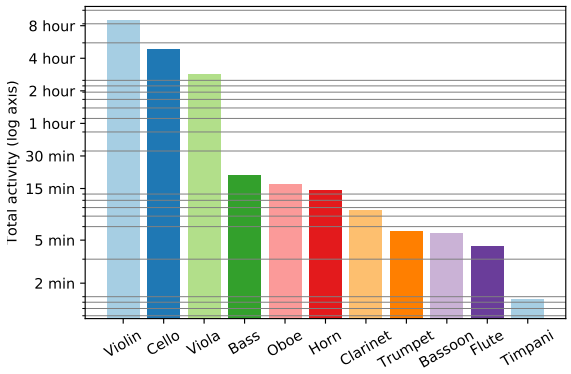


Figure 2. Instrument wise activity duration in EnsembleSet

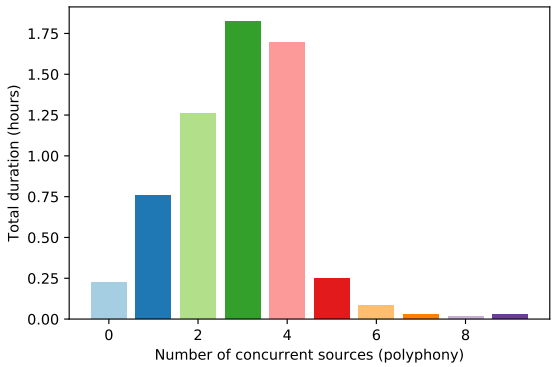


Figure 3. Polyphony distribution across EnsembleSet

articulation generation pipeline as the RWC sourced files. For some other files where the errors were minor (incorrect timing and track assignments), we used their corresponding MIDI files from the library to manually inspect and correct these MusicXML files.

3.2 BBC Symphony Orchestra Sample Library

This library was developed in partnership between BBC Studios and Spitfire Audio, by capturing a full orchestra as sections as well as individual section leaders. Each instrument was recorded for each note in a variety of articulation modes using multiple microphones placed at different positions in the room. For shorter notes, multiple iterations were recorded which are rendered in a round-robin fashion to simulate microtiming variations of real performers. The sample library was recorded in the same fashion as a film score would be recorded in a studio with a multi-microphone setup that enables the capture of each performer from different perspectives in the room. This is what allows us to simulate high quality recordings of chamber ensemble pieces from digital music scores, rendered using individual section leaders.

No.	Render Name	Type	# Mics	Pan
1	Mono	Bidirectional	1	Mono
2	Leader	Unidirectional	1	Stage Pan
3	Decca Tree	Omnidirectional	3	Stereo
4	Outriggers	Omnidirectional	2	Stereo
5	Ambient	Omnidirectional	2	Stereo
6	Balcony	Omnidirectional	2	Stereo
7	Stereo Pair	Coles 4038	2	Stereo
8	Mids	Omnidirectional	2	Stereo
9	Sides	Omnidirectional	2	Stereo
10	Atmos Front	Omnidirectional	2	Stereo
11	Atmos Rear	Omnidirectional	2	Stereo
12	Close	Unidirectional	1	Stage Pan
13	Close Wide	Unidirectional	1	Mono
14	Spill String	Unidirectional	15	Stage Pan
15	Spill Brass	Unidirectional	11	Stage Pan
16	Spill Woodwind	Unidirectional	12	Stage Pan
17	Spill Percussion	Unidirectional	10	Stage Pan
18	Spill Full	Unidirectional	48	Stage Pan
19	Mix 1	Mix	12	Stage Pan
20	Mix 2	Mix + FX	12	Stage Pan

Table 1. List of available renders in EnsembleSet. It must be noted that the Leader microphone is only available for string instruments.

3.2.1 Microphone Renders

The BBCSO sample library provides an entire portfolio of recording stems/microphones that a mix engineer would rely on while producing orchestral scores including traditional mixing setups like decca tree, outriggers, ambient microphones, far balcony mics, side mics and more modern setups including Atmos front and back mics placed at a height in the room. The placement of the individual microphones and each performer is shown in Figure 1. These recorded samples not only preserve the timbral changes that occur for a source recorded at various positions based on their distance, microphone type and directionality, but also preserves the phase shifts that occur across these different microphones placed at different distances w.r.t. the sources. The recorded samples are rendered without any time-correction, which implies that different mic renders would have different time delays and phase shifts based on the distance between the source and the mic. The renders also realistically reflect the timing adjustments performers for different sections make based on their instruments dynamics and position on stage.

It must be noted that in EnsembleSet, the positions for the close microphones are unique for each instrument, but the remaining room microphones are common across all instruments. Thus to simulate a realistic microphone bleed scenario, one can simply render each source at any given room microphone and downmix the resulting instrument stems. On the other hand, downmixing the close microphones would simulate the more typical scenario of music separation from a mixed song. Further details about individual microphone/mix setups can be found in Table 1.

3.2.2 Mixes

Apart from the individual microphone stems, the plugin also provides two professionally mixed stems. Mix 1 is a

Switch	Strings	Horn & Trumpet	Flute & Clarinet	Oboe & Bassoon
C-1	Legato	Legato	Legato	Legato
C#-1	Long	Long	Long	Long
D-1	Long Con Sordino	Staccatissimo	Trill Major 2nd	Trill Major 2nd
D#-1	Long Flautando	Marcato	Trill Minor 2nd	Trill Minor 2nd
E-1	Spiccato	Long Cuivre	Staccatissimo	Staccatissimo
F-1	Staccato	Long Sforzando	Tenuto	Tenuto
F#-1	Pizzicato	Long Flutter	Marcato	Marcato
G-1	Col Legno	Multi-tongue	Long Flutter	Multi-tongue
G#-1	Tremolo	Trill Major 2nd	Multi-tongue	-
A-1	Trill Major 2nd	Trill Minor 2nd	-	-
A#-1	Trill Minor 2nd	Long (muted)	-	-
B-1	Long Sul Tasto	Staccatissimo (muted)	-	-
C0	Long Harmonics	Marcato (muted)	-	-
C#0	Short Harmonics	-	-	-
D0	Bartok Pizzicato	-	-	-
D#0	Marcato	-	-	-

Table 2. List of keyswitch-articulation mappings for different instruments.

general starting point for a mix engineer with a good balance of the commonly used microphones like Decca Tree, Outriggers, Ambient, Balcony, Mids and Close mics. Mix 2 provides a more intense sound with some added compression, EQ and reverb. These stems are ideal to simulate the typical music separation scenario as the mixes provided present a good simulation of an unmastered and a mastered mix for an orchestral ensemble.

3.3 Articulation Automation

The BBCSO plugin allows rendering each note in a variety of articulations that are particular to each instrument. We use Dorico which in case of importing scores as MusicXML files, is capable of mapping articulations from MusicXML to keyswitches in the -1 octave in MIDI. Alternatively if articulations are unavailable, as is the case for importing scores as MIDI files, Dorico automatically selects either staccato or long articulation based on individual note lengths with a crossover at 187.5ms (16th note at 80bpm). The list of keyswitches and articulation mappings for each of the instruments available in EnsembleSet is shown in Table 2.

3.4 Dataset Contents

EnsembleSet contains a total of 6 hours and 9 minutes of multi-instrument, multi-mic data and is available on Zenodo [3] . The resulting total active duration of each instrument in EnsembleSet can be seen in Figure 2. The dataset presented is focused around string ensembles, and each of the 80 tracks presented in the dataset contains at least one string instrument, while the majority of pieces comprise string quartets. EnsembleSet also contains other woodwind and brass instruments, although their distribution is rather sparse. The overall polyphony distribution across the dataset is shown in Figure 3. Each song is also paired with its accompanying MIDI file which was used to generate the renders, and also contains the articulation information. Our implemented data preprocessing (described in section 4.2), data augmentation pipeline and other metadata related to the tracks such as song title, author, instrumentation and audio examples are available online [4] .

3.5 Limitations

While we have tried our best to make the synthesised recordings sound as realistic as possible, the achieved quality was still limited by the amount of information available in the source MIDI/lilypond files. All of the 9 tracks sourced from the RWC database have very good dynamics and realistic tempo variations in the renders, but since the source data was MIDI, the articulations are limited to long, staccato and pizzicato. For the 71 songs sourced from Mutopia, we were able to render from MusicXML for 30 of them, thus these are the only songs that are able to map to all possible articulations present in the source sheet music. For the remaining 41 tracks which were rendered from MIDI, the articulations are similarly limited to long, staccato and pizzicato. For all the songs sourced from Mutopia, the dynamics mapping available was limited due to limitations of the source lilypond format, thus resulting in each note having only one of 3 levels of velocity. While the instrument names in the renders have been standardised across the dataset, the accompanying MIDI files provided with each of the renders do not have standardised track names as they were preserved from the original track names from the source MIDI/Lilypond files.

4. EXPERIMENTS

To exhibit the value of our synthesised dataset, we use EnsembleSet to train a source separation model that is able to separate any chamber ensemble duet as explored in [30]. While we are training our model exclusively on our generated data, we evaluate on real-world data from the URMP dataset [20]. We make use of the multi-mic renders that are available in EnsembleSet as a form of data augmentation by randomising the mix/mic(s) presented to the

[3] https://zenodo.org/record/6519024

[4] http://c4dm.eecs.qmul.ac.uk/EnsembleSet/

Model	Train	Eval	SDR	SI-SDR
MSI [30]	URMP	URMP	+6.33 dB	-
DPTNet	URMP	ES	+6.29 dB	+4.37 dB
DPTNet	ES	URMP	+11.37 dB	+9.06 dB
DPTNet	ES	ES	+14.17 dB	+12.87 dB

Table 3. 2-source Chamber Ensemble Separation results.

model at each epoch. In addition, we use other augmentations including pitch shift and gain modulation to help the model generalise better to unseen source/microphone configurations. We utilise the same architecture as presented in [11] which is based on [10], modified to accommodate for 44.1kHz sample rate audio.

4.1 Model

We utilise the Dual-path Transformer (DPTNet) [10] based architecture using PIT [27] and modify the filterbank, scheduler and other network parameters to accommodate input segments at a sampling rate of 44.1kHz. Our model takes 2.97 second input frames (131072 samples) with 8 repeating separator units. We define the 1-D encoder filterbank to have a filter length of 32 samples with a hop size of 4 samples which resulted in best results in our experiments. Utilising a PIT loss for monotimbral ensembles is particularly well suited, as this enables our model to be able to separate any two monotimbral instruments regardless of their class.

4.2 Data

We train the model using all possible combinations of chamber ensemble duets playing simultaneously from EnsembleSet (ES) amounting to about 53 hours of data. To achieve this we implemented a novel dataloader that measures instrument activity confidence for each instrument track and identifies pairs of instrument segments where both the sources have some overlapping activity in all possible combinations (for eg: a string quartet piece for 2 source separation can be used as 6 different pairs of string duets). We used the URMP dataset (URMP) [20] to generate real examples for cross-validation and testing in a similar fashion resulting in 4.5 hours of 2 source mixtures. We utilise torch-audiomentations [34,35] for data augmentation such as gain modulation, channel swap and pitch-shifting by up to +/- 2 semitones. We also use the multi-mic renders of each instrument track as data augmentation by randomly choosing one of the 20 renders for each instrument for each iteration. It must also be noted that we maintain temporal and harmonic integrity of the mixtures through all the data augmentations. This is unlike the typical music separation data augmentation pipeline where the constituent parts of the mixtures are randomised across different songs at every epoch during training [36].

4.3 Training

We train the models for 100 epochs with early stopping patience of 10 epochs. We start with a learning rate of $5 \times e^{-3}$ with a scheduler that halves the learning rate if the validation loss does not improve for 3 epochs. We train the models on 4 x NVIDIA A100 GPUs using a distributed data parallel back-end. Each epoch in our experiments took 40 minutes with a batch size of 1 per GPU.

5. RESULTS

We present our baseline results based on the experiments described above and compare it to previous experiments conducted for a similar task as described in [30]. The results from [30] are based on a zero-shot learning + multi-task source informed (MSI) separation model designed to tackle the limitation of a very small training dataset. We compare our model's cross-dataset evaluation performance between the URMP Dataset [20] and EnsembleSet (ES) with the experiments from [30] as shown in Table 4.1. We find that our model trained on URMP and tested on ES reports similar separation quality as the MSI experiments from [30], although the test sets were not identical. The same model trained on ES and tested on URMP reports an improvement of 5dB in separation quality.

6. CONCLUSION

In this paper we introduced a new dataset constructed using digitised chamber ensemble scores and a professional orchestral sample library to address the lack of multitrack chamber ensemble datasets. We described our data generation process and data augmentation methods to enable generalisable deep learning solutions using the same. We provided a baseline for the task of separating 2 monotimbral instruments playing simultaneously and are able to show that models trained exclusively on our synthesised dataset are able to generalise to real world data for the same task. This outcome emphasises the strong dependence of the performance of deep learning on training regimes, in particular the quality of the training dataset.

The presented dataset not only contains high quality multi-microphone renders of various instruments, but is also accompanied by the MIDI files that were utilised for generating this dataset. This paired data can be utilised for various tasks including multi-instrument transcription [12], instrument recognition [13], score-informed source separation [13], microphone simulation [15], and automatic mixing [16].

While PIT is well suited for monotimbral ensemble separation as it can separate any 2 sources regardless of their instrument class, it is limited by polyphony where a model only works for mixtures with a fixed number of sources. In the future we intend to explore source conditioned separation models which would enable separating any particular source from a mixture. Although the efficacy of such solutions in the case of mixtures with multiple instances of same instruments has to be tested. In our current work we perform the separation on single channel audio, but we would like to extend our model to be capable of handling multi-channel audio input and utilise spatial information implicitly during separation.

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